Response to Reviewers

Manuscript ID: SOMED-26-RW01
Journal: *Scholarly Open: Medicine*
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Manuscript Title:
Prevent Earlier, Recognize Sooner, Treat Faster: A Proposed Evidence-Based Medicine–Integrated Rapid Response Framework (EBM-IRRF) for Acute Aortic Dissection

Dear Editor and Reviewers,

Thank you for the careful evaluation of the manuscript and the constructive recommendations. The reviewers' comments have provided valuable guidance for strengthening the article's scientific foundation, methodological transparency, clinical relevance, and organization.

The manuscript has been substantially revised to clarify its classification as a Perspective Article, explain the application of Systems-Based Scientific Reasoning (SBSR), expand the supporting scientific evidence, and distinguish established clinical recommendations from the proposed Evidence-Based Medicine–Integrated Rapid Response Framework (EBM-IRRF). Particular attention has been directed toward regional aortic networks, interhospital transfer, multidisciplinary coordination, organizational accountability, and the responsible application of emerging technologies.

The revised article uses a targeted narrative literature review and thematic evidence synthesis to inform conceptual framework development. It does not claim to be a systematic review, meta-analysis, original clinical investigation, or prospective validation study.

The following responses describe the revisions undertaken to address the reviewers' recommendations. An additional request concerning the manuscript's reader-oriented formatting is respectfully submitted for editorial consideration.

Response to Reviewer 1

Editorial Recommendation: Re-write and Re-submit

Comment 1. The manuscript should be more concise and clearly organized.

Response: Thank you for this recommendation. The manuscript has been reorganized into six major sections: Introduction, Literature Review, Materials and Methods, Results, Discussion, and Conclusions.

Repetitive material has been reduced, and the distinctions among scientific evidence synthesis, conceptual framework development, and interpretation have been clarified.

The revised structure is intended to improve scientific coherence while preserving the interdisciplinary scope necessary to examine the clinical and organizational challenges associated with acute aortic dissection (AAD).

Revised sections: Sections 1–6.

Comment 2. The manuscript requires additional scientific references.

Response: The literature review has been expanded to incorporate additional clinical guidelines, registry investigations, systematic reviews, observational studies, and relevant methodological literature.

Particular attention has been given to contemporary evidence concerning AAD prevention, diagnostic evaluation, treatment, regionalization, institutional procedural volume, interhospital transfer, multidisciplinary coordination, and emerging artificial intelligence (AI) applications.

References published during 2021–2026 have been prioritized where appropriate, while earlier foundational publications have been retained when essential to the scientific and historical context.

The reference list contains 40 entries. The author has reviewed the referenced publications and corrected an identified DOI mismatch involving the FUTURE-AI consensus guideline.

Revised sections: Sections 1–5 and References.

Comment 3. The evolution of AAD management should be discussed.

Response: A dedicated subsection has been developed to examine major developments in AAD diagnosis and treatment.

The discussion addresses advances in cardiovascular imaging, surgical management, endovascular interventions, evidence-based clinical guidelines, specialized aortic centers, and regional emergency-care networks.

Table 1 summarizes major developments in AAD management and their clinical implications without assuming that each innovation independently produced improvements in mortality or other patient outcomes.

Revised sections: Section 2.6 and Table 1.

Comment 4. Tables should be added to improve clarity.

Response: Five tables have been incorporated to improve the organization, accessibility, and interpretation of the scientific evidence and proposed framework.

The tables summarize major developments in AAD management, distinguish established clinical evidence from proposed organizational strategies, describe the integrated EBM-IRRF components, and identify potential indicators for future evaluation.

The tables are intended to support critical interpretation rather than present conceptual proposals as established clinical findings.

Revised sections: Relevant sections and Tables 1–5.

Comment 5. Figures should be added to illustrate the proposed framework.

Response: A conceptual illustration of EBM-IRRF has been developed to clarify the relationships among its seven interconnected components and their integration across the AAD care continuum.

The framework includes four clinical and coordination components supported by three cross-cutting enabling functions: continuous evidence-based medicine (EBM), appropriately validated and clinician-supervised AI, and organizational leadership, accountability, and continuous quality improvement.

The illustration presents a proposed organizational and clinical integration model rather than a validated clinical pathway or intervention.

Revised section: Section 4 and Figure 1.

Response to Reviewer 2

Editorial Recommendation: Re-review and Accept With Major Changes

Comment 1. The manuscript's methodology and article type require clarification.

Response: Thank you for identifying the need for greater methodological transparency. The manuscript has been explicitly classified as a **Perspective Article** presenting an evidence-informed conceptual framework.

The Materials and Methods section has been revised to explain the application of **Systems-Based Scientific Reasoning (SBSR)**, an author-defined integrative conceptual reasoning approach informed by established scientific methodologies, clinical experience, EBM, systems science, critical reasoning, and implementation science.

SBSR was applied through a targeted narrative literature review, thematic evidence synthesis, and conceptual framework development to examine nine interconnected scientific and clinical questions concerning AAD.

The revised manuscript distinguishes SBSR from an independently validated research methodology. It also clarifies that the literature review was not conducted as a systematic review or meta-analysis and that no original patient-level research or statistical analysis was performed.

The literature search strategy, evidence selection considerations, and synthesis procedures are described to improve methodological transparency.

Revised sections: Abstract, Section 1.5, and Sections 3.1–3.4.

Comment 2. The distinction between established evidence and the proposed framework should be clarified.

Response: The revised manuscript more clearly distinguishes three categories of scientific information:

1. Established clinical practices supported by guidelines and published evidence.
2. Evidence-informed organizational and healthcare delivery strategies.
3. Proposed conceptual innovations requiring further scientific evaluation.

EBM-IRRF is presented as an evidence-informed conceptual framework rather than an implemented or clinically validated intervention.

Statements concerning its potential benefits have been qualified to avoid implying demonstrated improvements in diagnostic timeliness, clinical outcomes, patient safety, or organizational performance.

The revised discussion emphasizes that feasibility, safety, effectiveness, incremental benefits, and generalizability require independent prospective evaluation.

Revised sections: Sections 3–6.

Comment 3. Additional evidence concerning regionalization and interhospital transfer is needed.

Response: The literature review has been expanded to incorporate evidence concerning specialized aortic centers, institutional procedural volume, interhospital transfer, regional aortic networks, and structured emergency-response pathways.

The revised discussion examines the potential advantages of regionalized AAD care while recognizing the limitations of the supporting evidence.

Particular attention has been given to observational study designs, patient selection, institutional differences, transfer delays, and the need to balance rapid access to specialized care against the risks associated with delayed definitive treatment.

The manuscript does not assume that regionalization is universally beneficial under all circumstances. Instead, it identifies regional coordination as an important area for context-specific implementation and further evaluation.

Revised sections: Sections 2.7, 4, and 5.

Comment 4. Multidisciplinary coordination and organizational leadership require further explanation.

Response: The proposed framework has been clarified to distinguish clinical decision-making authority from organizational coordination and accountability.

Qualified treating physicians retain responsibility for clinical assessment and treatment decisions, while organizational leaders support emergency readiness, resource mobilization, communication, interdepartmental coordination, and continuous quality improvement.

The manuscript also addresses the potential roles of multidisciplinary teams, specialized aortic centers, and regional referral networks.

Importantly, the revised framework emphasizes that additional consultation, administrative escalation, or organizational review must not delay clinically indicated emergency treatment.

Revised sections: Sections 4 and 5.

Comment 5. The proposed role of artificial intelligence requires clarification.

Response: The revised manuscript presents AI as a potential supplementary clinical and operational resource rather than an autonomous decision-maker.

The discussion addresses AI applications in diagnostic support, imaging interpretation, risk assessment, clinical workflow coordination, and healthcare operations.

The manuscript emphasizes that clinical AI applications require appropriate validation, qualified clinician oversight, responsible data governance, patient safety safeguards, and continuous performance monitoring.

Relevant contemporary literature, including the 2025 FUTURE-AI international consensus guideline, has been incorporated to support the discussion of trustworthy and deployable AI in healthcare.

The manuscript also distinguishes general healthcare AI governance principles from AAD-specific evidence of clinical effectiveness.

Revised sections: Sections 2.10, 4, and 5.

Comment 6. The framework requires clearer evaluation criteria.

Response: The revised manuscript identifies potential clinical and organizational indicators for evaluating EBM-IRRF.

These include diagnostic timeliness, specialist response, emergency activation, interhospital transfer intervals, time to definitive treatment, major complications, mortality, surveillance adherence, patient-centered outcomes, organizational performance, and unintended consequences.

Future investigations are proposed to evaluate the framework's feasibility, implementation fidelity, safety, effectiveness, and potential incremental benefits compared with existing clinical and organizational practices.

Suggested research approaches include feasibility assessments, prospective multicenter investigations, comparative implementation studies, and evaluation of individual framework components.

The manuscript acknowledges that these proposed evaluation indicators have not yet been tested as an integrated EBM-IRRF measurement system.

Revised sections: Sections 4.5 and 5.6; Table 5.

Comment 7. Ethics approval and informed consent should be clarified.

Response: The ethics declaration has been clarified to reflect the manuscript's classification as a Perspective Article.

The revised statement reads:

Ethics Approval and Informed Consent: This Perspective Article synthesizes previously published scientific evidence and proposes a conceptual framework. No original human participant research, patient-level data collection, or identifiable patient information was involved. Accordingly, institutional review board approval and informed consent were not required.

Revised section: Ethics Approval and Informed Consent declaration.

Comment 8. Unsupported patient-specific statements should be addressed.

Response: The reference to Senator Lindsey Graham has been removed from the revised manuscript.

The discussion now focuses on published scientific evidence, generalizable clinical challenges, and healthcare delivery considerations rather than unsupported assumptions concerning an identifiable individual's medical history or clinical management.

This revision is intended to strengthen scientific accuracy, clinical relevance, and professional objectivity.

Revised sections: Introduction and relevant clinical discussion.

Additional Request for Editorial Consideration

Retaining Bold Opening Sentences to Improve Scientific Readability

The author respectfully requests permission to retain the bold formatting of the opening sentence in each paragraph as an intentional, reader-oriented scientific communication strategy.

This formatting approach has been incorporated throughout the revised manuscript to highlight the principal scientific message of each paragraph. Its purpose is to facilitate efficient reading, improve the identification of important concepts, and help readers navigate the integration of clinical evidence, EBM, healthcare operations, and organizational accountability.

The proposed approach is intended to improve the accessibility and practical usefulness of complex scientific information without compromising academic rigor. Clinicians, researchers, healthcare administrators, and other professionals often need to identify relevant findings and practical implications efficiently.

By emphasizing the opening sentence, the manuscript provides an additional level of visual organization while retaining the supporting scientific explanations, qualifications, and citations.

The author recognizes that this approach differs from conventional scientific manuscript formatting and may not be explicitly addressed by AMA style conventions or the journal's editorial guidelines. Nevertheless, the author considers it a potentially innovative and practical approach to scientific communication, particularly for interdisciplinary articles intended for both clinical and organizational audiences.

The approach is proposed as a presentation feature rather than an established or empirically validated method for improving scientific comprehension.

The author's strong preference is to preserve the bold opening sentence of each paragraph in both the accepted manuscript and the final published article. This request is made to improve readability, knowledge translation, and practical accessibility rather than to depart from established standards of scientific accuracy, transparency, or professionalism.

If the proposed formatting raises concerns regarding journal consistency, editorial presentation, or production requirements, the author would welcome discussion of reasonable alternatives.

A possible compromise would be to retain bold opening sentences in selected sections, particularly the Introduction, Discussion, and Conclusions, where rapid identification of principal arguments may be especially valuable.

The author respectfully requests that the editorial team evaluate this formatting feature on its practical merits and, if appropriate, consult the reviewers before making a final determination.

The author remains willing to comply with mandatory journal formatting requirements while expressing a strong preference for retaining this reader-friendly presentation wherever editorially permissible.

Summary of Major Revisions

The revised manuscript incorporates substantive changes intended to address the reviewers' scientific, methodological, organizational, and presentation concerns.

1. **Improved article organization:** Reorganized the manuscript into six major sections and reduced unnecessary repetition.
2. **Clarified article classification:** Explicitly identified the manuscript as a Perspective Article rather than a systematic review, meta-analysis, or original clinical investigation.
3. **Strengthened methodological transparency:** Defined SBSR as an author-developed integrative conceptual reasoning approach and clarified its application through targeted narrative evidence synthesis and conceptual framework development.
4. **Expanded scientific evidence:** Incorporated contemporary clinical and organizational literature, with particular emphasis on regionalization, interhospital transfer, institutional procedural volume, and multidisciplinary coordination.
5. **Improved scientific presentation:** Added five tables and a conceptual figure to clarify the evidence base, framework components, and proposed evaluation indicators.
6. **Clarified EBM-IRRF:** Distinguished its four clinical and coordination components from three cross-cutting enabling functions.
7. **Distinguished evidence from proposals:** Clarified that EBM-IRRF is a proposed framework whose clinical effectiveness and organizational benefits remain unvalidated.
8. **Strengthened organizational accountability:** Distinguished clinical decision-making authority from administrative coordination, emergency readiness, and quality improvement responsibilities.
9. **Clarified AI governance:** Emphasized validation, clinician oversight, data governance, patient safety, and continuous performance monitoring, supported by relevant contemporary literature.
10. **Expanded future research recommendations:** Identified potential indicators and study designs for evaluating feasibility, safety, effectiveness, implementation, and generalizability.

11. **Clarified ethical considerations:** Confirmed that the article does not involve original human participant research or identifiable patient information and removed unsupported patient-specific statements.
12. **Reviewed reference accuracy:** Reviewed the 40 cited publications and corrected an identified DOI mismatch involving the FUTURE-AI consensus guideline.
13. **Requested editorial consideration of formatting:** Provided a rationale for retaining bold opening sentences as an intentional reader-oriented feature intended to improve scientific communication.

Closing Statement

We sincerely appreciate the reviewers' thoughtful recommendations, which have contributed to strengthening the manuscript's scientific clarity, clinical relevance, methodological transparency, and organizational coherence.

The revised article emphasizes that EBM-IRRF is a proposed conceptual framework intended to complement established clinical guidelines and emergency-care systems rather than replace them. Its principal contribution is the conceptual integration of evidence-based clinical practice with coordinated healthcare operations across prevention, recognition, referral, treatment, and continuous quality improvement.

Systems-Based Scientific Reasoning (SBSR) is presented as the author's integrative conceptual reasoning approach rather than an independently validated scientific methodology. Its application informed the examination of nine interconnected scientific and clinical questions and the development of EBM-IRRF.

The effectiveness of EBM-IRRF remains to be established through independent prospective clinical and organizational evaluation. Accordingly, the revised manuscript distinguishes established evidence from proposed innovations and identifies measurable outcomes for future investigation.

In addition to addressing the reviewers' scientific and methodological recommendations, the author respectfully seeks editorial consideration of the manuscript's bold opening sentences as a potentially useful innovation in scientific communication.

This presentation approach is intended to facilitate efficient reading, improve the identification of principal scientific arguments, and strengthen the practical accessibility of interdisciplinary research.

The author respectfully submits the revised manuscript for reconsideration and welcomes further editorial recommendations.

Sincerely,

Sam Lee, MD, PhD, DHA Candidate
Founder, President & Chief Clinical EBM Scientist
Applied Clinical EBM Institute (ACEI)
Honolulu, Hawai‘i, USA